

1. 证明极坐标下的柯西黎曼条件是: $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$.

1.

$$\text{设 } z = x + yi = re^{i\theta} \text{ 则 } x = r \cos \theta, y = r \sin \theta, f(z) = u + iv$$

$$\text{从而 } u_r = u_x \cos \theta + u_y \sin \theta, u_\theta = -u_x r \sin \theta + u_y r \cos \theta$$

$$v_r = v_x \cos \theta + v_y \sin \theta, v_\theta = -v_x r \sin \theta + v_y r \cos \theta$$

$$\text{由 } u_x = v_y, u_y = -v_x$$

$$\text{即得 } u_r = \frac{1}{r} v_\theta, v_r = -\frac{1}{r} u_\theta$$

2. 三角函数与双曲函数 由(2.9), 当 $x=0$ 时推得

2. 试利用柯西黎曼条件, 证明 $\sin \bar{z}$ 不是解析函数.

2.

$$\begin{aligned} \sin z &= \sin(x+yi) = \sin x \cos(yi) + \cos x \sin(yi) \\ &= \sin x \cosh y + i \cos x \sinh y \end{aligned}$$

$$\sin \bar{z} = \sin x \cosh y - i \cos x \sinh y$$

$$\Rightarrow u = \sin x \cosh y, v = -\cos x \sinh y$$

$$u_x = \cos x \cosh y, v_y = -\cos x \cosh y$$

只有当 $x = 2k\pi \pm \frac{\pi}{2} (k=0, \pm 1, \pm 2, \dots)$ 时才有 $u_x = v_y$
 $\sin \bar{z}$ 在复平面上不解析.

3. 问下列各函数有哪些奇点, 并确定奇点的类别:

$$(1) \frac{1}{\sin z - \sin 1}$$

$$3. (1) \text{ 由 } Q(z) = \sin z - \sin 1 = 2 \cos \frac{z+1}{2} \sin \frac{z-1}{2} = 0.$$

$$z_1 = 2k\pi + \pi - 1, z_2 = 2k\pi + 1 (k \in \mathbb{N}).$$

$$Q'(z_i) = \mp \cos 1 (i=1, 2), \neq 0$$

故 z_i 为一阶极点, $z=\infty$ 为以上极点的极限点,
 故为非孤立奇点.

$$(2) \sin \frac{1}{1-z}$$

$$(2) \text{ 由 } \sin\left(\frac{1}{z-a}\right) = \frac{1}{z-a} - \frac{1}{(z-a)^3 3!} + \frac{1}{(z-a)^5 5!} + \dots$$

$$0 < |z-a| < +\infty$$

不存在有限的 $m \in \mathbb{N}$ $\lim_{z \rightarrow a} (z-a)^m f(z) = k$, k 为非0常数

故 $z=a$ 为本质奇点.

$$(3) \frac{z-1}{z(z^2+4)^2}$$

(3) 令 $z(z^2+4)=0$ 得 $z=0$ 及 $z=\pm 2i$ 分别是 $z(z^2+4)=0$ 的一级和二级零点, 从而分别是原函数的一级和二级极点.

又 $\frac{z-1}{z(z^2+4)} \rightarrow 0 (z \rightarrow \infty)$, 所以 $z=\infty$ 为可去奇点.

4. 将下列函数展开为相应环区域的洛朗级数

$$(1) \frac{1}{(z-1)(z-2)}, (|z| < 1 \text{ 和 } 1 < |z| < 2)$$

$$(1) f(z) = \frac{1}{z-2} - \frac{1}{z-1}$$

$$\textcircled{1} \text{ 在 } |z| < 1 \text{ 内 } |z| < 2 \quad \left|\frac{z}{2}\right| < 1. \text{ 利用 } \frac{1}{1-u} = \sum_{n=0}^{\infty} u^n \quad (|u| < 1)$$

$$f(z) = \frac{1}{1-z} - \frac{1}{2(1-\frac{z}{2})} = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) z^n$$

$$\textcircled{2} \text{ 在 } 1 < |z| < 2 \text{ 内 } \left|\frac{1}{z}\right| < 1, \left|\frac{z}{2}\right| < 1$$

$$f(z) = -\frac{1}{z} \cdot \frac{1}{1-\frac{z}{2}} - \frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\frac{1}{z} \sum_{n=0}^{\infty} \frac{z^n}{2^n} - \frac{1}{z} \sum_{n=1}^{\infty} \frac{1}{z^{n-1}}$$

$$= -\sum_{n=0}^{\infty} \frac{z^{n-1}}{2^n} - \sum_{n=1}^{\infty} \frac{1}{z^n}$$

$$(2) \frac{z}{z^2(z-i)}, (0 < |z-i| < 1 \text{ 和 } 1 < |z-i| < +\infty)$$

$$\begin{aligned}
 & \text{für } 0 < |z-i| < 1 \\
 \frac{z}{z^2(z-i)} &= \frac{1}{z(z-i)} = \frac{1}{(z-i+i)} \frac{1}{z-i} \\
 &= \frac{1}{i} \frac{1}{1 + \frac{z-i}{i}} \frac{1}{z-i} = (-1) \frac{1}{1 - \left(\frac{z-i}{i}\right) \cdot (-1)} \frac{1}{z-i} \\
 &= (-1) \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-i}{i}\right)^n \cdot \frac{1}{z-i} = \sum_{n=0}^{\infty} (-1)^{n+1} \left(\frac{z-i}{i}\right)^{n+1}
 \end{aligned}$$

$$\text{für } 1 < |z-i| < +\infty \quad \left| \frac{i}{z-i} \right| < 1$$

$$\begin{aligned}
 \frac{1}{z(z-i)} &= \frac{1}{(z-i)} \frac{1}{z-i+i} = \frac{1}{(z-i)} \frac{1}{1 + \frac{i}{z-i}} \frac{1}{z-i} \\
 &= \left(\frac{1}{z-i}\right)^2 \frac{1}{1 - \frac{i}{z-i} \cdot (-1)} = \left(\frac{1}{z-i}\right)^2 \sum_{n=0}^{\infty} (-1)^n \left(\frac{i}{z-i}\right)^n \\
 &= - \sum_{n=0}^{\infty} (-1)^n \left(\frac{i}{z-i}\right)^{n+2}
 \end{aligned}$$